

ASTR 210 Fall 2025 — Homework set 10
50 points plus optional extra credit
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Solutions need to show all important intermediate calculations, diagrams, and explanations in addition to the final answer to receive full credit. Please check the legibility of homework solutions uploaded to the class website as we cannot grade or give credit for illegible or unreadable work. Units do not need to be specified if an expression or equation is requested; for numerical calculations please use the specified units for the answer, or if not specified, SI or cgs units. Please use Table A.1 for the value of physical constants and Table A.2 for the value of astronomical constants, unless otherwise specified in the problem. Please use the correct number of significant figures.

10.1 Supernova!

Supernova 1987a exploded in the Large Magellanic Cloud, a dwarf galaxy near to the Milky Way, and was observed (as you might guess from the name) in early 1987. It was a big deal to astronomers, as nobody had seen a supernova anywhere close by since 1604! It was also the first nearby supernova for which we had neutrino detectors. However, it turns out that there have been other supernovae in our galaxy that we just didn't notice. One such supernova that slipped under our collective radar is called G1.9+0.3. If you are deeply curious you can read more about it [here](#).

- (a) The shock wave of SNR G1.9+0.3 is measured to be roughly circular and 1.2 arcminutes in diameter, and to be expanding at $14,000 \text{ km s}^{-1}$. Assume that it has a distance of 8.5 kpc and it has been expanding at a constant velocity. Extrapolate backwards in time to find when it would have exploded, i.e. when it had a radius $R = 0$. [3 pts]
- (b) A typical supernova emits approximately 10^{57} neutrinos, which will diffuse out of the collapsing core over a timescale $\Delta t \sim 15 \text{ s}$. Assuming the neutrinos are emitted isotropically, calculate the neutrino flux (units: neutrinos per m^2 per second) that would be expected at Earth from SN G1.9+0.3. [3 pts]
- (c) At maximum luminosity, a typical supernova has a luminosity (in photons, not neutrinos) of $10^9 L_\odot$. That's pretty bright. However, there is also a lot of dust in the plane of the Milky Way galaxy, and that dust could have hidden the supernova from our view. To explore that possibility, work out the apparent magnitude m_V that the supernova would have had. Lines of sight towards the galactic center have extinctions as high as $A_V = 30$ magnitudes. Convert the bolometric luminosity $L_{\text{bol}} = 10^9 L_\odot$ to a bolometric magnitude M_{bol} and assume for simplicity that $M_V = M_{\text{bol}}$. Compare the apparent magnitude m_V to the limiting magnitude for naked-eye visibility, which is somewhere around $m_V = +6$ (chapter 13), and to that of large telescopes (around $m_V = +30$). You should find that it would be difficult to see a supernova behind all of that dust. [3 pts]

10.2 Stellar remnants and angular momentum

We know that the Sun should end up as a white dwarf, and a typical radius for a white dwarf is something like 6000 km. What would be the rotation period of the Sun if it collapsed to that radius while conserving angular momentum? (You can get the rotation period of the Sun in chapter 7 and there's some discussion of angular momentum conservation in chapter 17.) [4 pts]

10.3 The nature of pulsars

As your text describes, pulsars were first discovered as regular, repeating pulses of radio emission from sources that are otherwise difficult to detect. Setting aside questions related to how the radio emission is produced (it's not blackbody radiation), let's explore some models that might or might not be able to explain a pulsation period of 1.337 seconds.

- (a) Suppose the pulsations come from a binary star containing two white dwarf stars that each have a mass of $0.7 M_{\odot}$ and a radius of about 7×10^6 m (see equation 18.17). If the white dwarf stars are orbiting each other with a period of 1.337 seconds, what is the semi-major axis of their orbit? [3 pts]
- (b) Comment on whether the previous part gives a plausible explanation for a pulsar, and if not, why not. [2 pts]
- (c) Now suppose the pulsar is not a binary but is one rotating star with a hot spot. Show that a spinning white dwarf could produce a pulsation with period P if its rotation velocity (at its equator) is [3 pts]

$$v_{\text{rot}}(P, M) \approx 0.1c \left(\frac{P}{1 \text{ s}} \right)^{-1} \left(\frac{M}{0.7 M_{\odot}} \right)^{-1/3}.$$

Here c is the speed of light.

- (d) Fast rotation produces a centrifugal acceleration that can break up a spinning object. The maximum possible rotation speed for which an object can hold together by gravity is the critical breakup speed v_{crit} . Find an approximate expression for v_{crit} by assuming the star remains spherical and setting the centrifugal acceleration at the equator to be equal to the star's gravity g . You should find $v_{\text{crit}} = \sqrt{GM/R}$. [3 pts]
- (e) Use the properties of white dwarf stars to show that their breakup speed is also given by [4 pts]

$$v_{\text{crit}}(P, M) \approx 0.01c \left(\frac{M}{0.7 M_{\odot}} \right)^{2/3}.$$

- (f) Based on the above results, do you find that a rotating white dwarf star with a mass of about $0.7 M_{\odot}$ is a plausible model for pulsations with a period of 1.337 s? Why or why not? [2 pts]
- (g) Redo parts (c), (e), and (f) for neutron stars rather than white dwarf stars. Use your results to demonstrate that a rotating neutron star of mass $1.4 M_{\odot}$ provides a plausible model for pulsations with a period of 1.337 s. [5 pts]

10.4 Mass models of the Milky Way Galaxy

Figure 19.19 in the text shows a recent estimate of the rotation curve of the Galaxy. It shows that the rotation speed v_c or Θ peaks at about 255 km s^{-1} at a radius of 0.4 kpc and falls to a low of 192 km s^{-1} at 2.8 kpc. Then it rises to about 220 km s^{-1} at a radius of 8 kpc, where the Sun orbits, and remains roughly constant to 235 km s^{-1} at a radius of 17 kpc. From these data, estimate the mass $M(R)$ of the Galaxy enclosed within those four radii. It will help your readers if you make a small table with columns for the radius values, the velocities, and the masses enclosed within those radii. Quote the masses in units of $M_{\odot}$. [4 pts total]

10.5 Galactic center black hole and S0-2

The star S0-2, which has been observed orbiting the black hole at the center of our Galaxy, has an orbital semimajor axis of 960 AU, a period of 15.6 yr, and an orbital eccentricity of 0.867.

- (a) From these data, estimate the mass of the black hole at the center of our Galaxy. [3 pts]
- (b) What is the star's maximum speed around the black hole, and what is its minimum distance from the black hole? Quote the speed both in linear units (km s^{-1}) and as a fraction of the speed of light. [3 pts]
- (c) How close does the star get to the black hole's Roche limit? You'll need the mass and radius of the star itself, which you can estimate from Table A.5 knowing that the star's spectral class is B1 V. For simplicity, call it approximately B0 V. Comment on whether this star is in any danger of tidal disruption. [5 pts]

EXTRA CREDIT

10.6 What do those Oort constants actually mean?

The physical interpretations of the Oort constants A and B can be a little obscure. To help get a better understanding of their meanings, use their definitions to show that circular Keplerian orbits have

$$\frac{A - B}{A + B} = 2.$$

“Keplerian orbits” are those resulting from a point mass, like we have in the Solar system, where $M(r)$ is constant because virtually all the mass is at the center. Comment on whether the observed values of A and B in our Galaxy match the prediction for Keplerian orbits. [8 pts]